

All Chips In?

Commentary on the Bank of England's AI Financial Stability Report

December 2025

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The Bank of England's October 2025 report on AI-related asset valuations raises important questions about financial stability in an era of unprecedented technology investment. This commentary examines the report through three historical lenses: the AI winters of the 1970s–1990s, Charles Mackay's *Extraordinary Popular Delusions and the Madness of Crowds* (1841), and Carl Benedikt Frey's *The Technology Trap* (2019). The central finding is a factor-of-ten divergence in automation estimates that should give any observer pause-for-thought.

The BoE Report: Key Findings

The report is notable for its measured tone – it neither proclaims a bubble nor dismisses the risk. AI stocks now comprise roughly 44% of the S&P 500 market capitalisation, up from 26% in late 2022. The cyclically adjusted price-to-earnings ratio is close to its dot-com peak, with AI stocks trading at a median forward P/E of 31x versus 19x for the broader index.

The infrastructure buildout is staggering: McKinsey estimates \$5.2 trillion in AI data centre CAPEX by 2030. Crucially, Morgan Stanley projects \$2.9 trillion in AI infrastructure spending between 2025–2028, with **\$1.5 trillion from external financing, including \$800 billion from private credit**. The shift from internal cash flows to debt financing transforms the risk profile entirely.

The authors note that Bank of Italy staff have shown these valuations are justifiable *if* projected earnings growth materialises. This conditional is the crux of the matter.

The 10x Estimation Divergence

The single most important data point for assessing bubble risk is the staggering gap between authoritative automation forecasts.

The 50% figure: The Bank of England's chief economist Andy Haldane has warned that machines could take over 80 million American and 15 million British jobs over the next 10–20 years – roughly half the workforce. 'These machines are different,' he argued. 'Unlike in the past, they have the potential to substitute for human brains as well as hands.'

The 5% figure: Nobel laureate Daron Acemoglu calculates that only 5% of jobs are ripe to be automated over the next decade. His analysis: while 20% of tasks *could* theoretically be automated, only a quarter of those—5% economy-wide—can be *profitably* automated. 'You're not going to get an economic revolution out of that 5%,' he notes. He expects AI to increase GDP by merely 1.1–1.6% over ten years.

The methodological divide is crucial. Haldane assesses *technical feasibility*; Acemoglu asks about *profitable automation*. A 2024 MIT FutureTech study found that only 23% of automatable computer vision tasks could be profitably implemented within ten years. Technical capability and economic viability are not the same thing.

If Haldane is right, current valuations may be conservative. **If Acemoglu is right**, \$5 trillion in infrastructure spending represents a colossal misallocation of capital.

Historical Precedents

The AI Winters

The term 'AI winter' appeared in 1984 when Roger Schank and Marvin Minsky warned that enthusiasm had spiralled out of control. They described a chain reaction: pessimism in the AI community, then the press, then severe funding cutbacks. Three years later the billion-dollar AI industry collapsed. Both winters were precipitated not by fraud, but by a gap between capability and expectation - precisely the dynamic the 50%/5% divergence suggests today.

The BoE report's discussion of the DeepSeek event is revealing: the introduction of an efficient reasoning model triggered falls across AI-impacted stocks. An algorithmic breakthrough challenging the assumption that massive compute is necessary can cause rapid revaluation – the kind of sentiment reversal that precipitated previous winters.

Mackay's Manias

Mackay's 1841 chronicle of the South Sea Bubble, Mississippi Scheme, and tulip mania remains relevant because the psychological mechanisms are invariant: 'Whole communities suddenly fix their minds upon one object, and go mad in its pursuit.' Yet modern researchers have shown Mackay exaggerated the tulip bubble's scale. Ironically, Mackay himself was caught in the Railway Mania of the 1840s, writing 'there is no reason whatever to fear a crash' months before the collapse. Even chroniclers of bubbles can be caught in them.

Frey's Technology Trap

Frey distinguishes 'labour-enabling' from 'labour-replacing' technologies. Acemoglu's critique aligns: 'We're using AI too much for automation and not enough for providing expertise and information to workers.' Current applications are predominantly labour-replacing (customer service, coding assistance), producing what Acemoglu calls 'so-so technology' – applications that perform only marginally better than humans but cost less. The deeper risk from Frey's framework is political: distributional consequences may generate instability that undermines technological progress itself.

What's Different This Time

The debt overhang: Previous AI winters were equity stories – funding disappeared but systemic contagion was limited. With \$800 billion in private credit projected for AI infrastructure, debt defaults could cascade through the financial system. Projects financed assuming 50% automation will face revenue shortfalls if only 5% materialises.

Physical path dependencies: The IEA estimates data centre electricity demand will more than double by 2030, exceeding Japan's consumption. Every megawatt requires 25–35 tonnes of copper. These aren't software projects that can be quietly abandoned.

Demonstrated capability: Unlike dot-com companies with no revenue, current AI firms generate substantial income. The question is whether growth trajectories justify valuations – a genuine judgment call. As Acemoglu notes, LLMs 'can do that in a few places with some human supervisory oversight—like coding—but in most places they cannot.'

Conclusion

The 50% versus 5% divergence is the central question for AI valuation. A 5% automation figure does not justify \$5 trillion in infrastructure spending. A 50% figure might. We do not know which estimate is correct.

That uncertainty makes the BoE's concerns about debt-financed AI infrastructure particularly pertinent. Equity losses are absorbed by shareholders. Debt defaults cascade through the financial system.

The BoE is right to flag the growing financial stability risks as debt financing increases. Whether current valuations constitute a bubble depends entirely on whether AI capabilities continue their rapid advancement and, crucially, whether that advancement can be monetised. The Bank of Italy's work suggests they could be justified. The AI winters suggest they might not be. Mackay suggests we won't know until after the fact.

Authors Note: I have consolidated 25 years worth of experience working in the AI / ML industry to bring together more recent commentary culminating in the BoE report. At a time where Schank, Minsky et al were amongst the godfathers of AI, followed by the Lighthill report and here we are now, we are still troubled by technology advances in a similar way to the productivity paradox.

References

Primary Sources

Bank of England (2025). *'All chips in! Would a fall in AI-related asset valuations have financial stability consequences?'* Bank Overground, 24 October 2025.

<https://www.bankofengland.co.uk/bank-overground/2025/all-chips-in-ai-related-asset-valuations-financial-stability-consequences>

Acemoglu, D. (2024). *'The Simple Macroeconomics of AI'*, NBER Working Paper No. 32487, May 2024.

<https://www.nber.org/papers/w32487>

Albori, M. et al. (forthcoming). Bank of Italy staff analysis on AI company valuations (cited in Bank of England, 2025).

Haldane, A. (2015–2018). Speeches on automation and employment, Bank of England.

<https://www.bankofengland.co.uk/speech/2019/andy-haldane-speech-at-glasgow-school-of-art>

Lighthill, J. (1973). *'Artificial Intelligence: A General Survey'*, Science Research Council, UK.

https://en.wikipedia.org/wiki/Lighthill_report

MIT FutureTech, Productivity Institute & IBM (2024). Study on profitable automation of computer vision tasks.

<https://www.csail.mit.edu/news/rethinking-ais-impact-mit-csail-study-reveals-economic-limits-job-automation>

Books

Frey, C.B. (2019). *The Technology Trap: Capital, Labor, and Power in the Age of Automation*. Princeton University Press.

<https://press.princeton.edu/books/hardcover/9780691172798/the-technology-trap>

Frey, C.B. & Osborne, M.A. (2017). 'The Future of Employment: How Susceptible Are Jobs to Computerisation?' *Technological Forecasting and Social Change*, 114, 254–280.

<https://www.sciencedirect.com/science/article/abs/pii/S0040162516302244>

Mackay, C. (1841). *Extraordinary Popular Delusions and the Madness of Crowds*. London: Richard Bentley.

<https://www.gutenberg.org/files/24518/24518-h/24518-h.htm>

Acemoglu, D. & Johnson, S. (2023). *Power and Progress: Our Thousand-Year Struggle Over Technology and Prosperity*. PublicAffairs.

<https://shapingwork.mit.edu/power-and-progress/>

Industry Reports

Goldman Sachs Research (2025). Data centre power demand projections.

International Energy Agency (2025). Global data centre electricity demand forecast.

McKinsey & Company (2025). AI data centre capital expenditure projections to 2030.

Morgan Stanley Research (2025). AI infrastructure financing projections 2025–2028.

Odlyzko, A. 'Charles Mackay's own extraordinary popular delusions and the Railway Mania'. University of Minnesota.

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1927396

Historical Sources on AI Winters

Schank, R. & Minsky, M. (1984). AAAI panel discussion on AI winter risks.

Haigh, T. (2023–2024). 'There Was No First AI Winter' and 'How the AI Boom Went Bust', *Communications of the ACM*.

<https://cacm.acm.org/opinion/there-was-no-first-ai-winter/>

Garber, P. & Goldgar, A. Research on the scale of Dutch tulip mania.